

Question:
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Key relation : For a particle moving with momentum p , the de Broglie wavelength is $\lambda = h/p$. For an electron, $p = \sqrt{2mE}$ because kinetic energy $K = p^2/(2m)$, so $\lambda = h/(\sqrt{2mE})$.

- Take $h = 6.625 \times 10^{-27}$ erg $\cdot$ s (units match with λ in cm), $m = 9.1 \times 10^{-28}$ g (electron mass), and $E = 1$ eV = 1.602×10^{-12} erg.
- Compute the momentum: $p = \sqrt{(2)(9.1 \times 10^{-28})(1.602 \times 10^{-12})} \approx 5.92 \times 10^{-19}$ erg $\cdot$ s.
- Then $\lambda = h/p \approx (6.625 \times 10^{-27}) / (5.92 \times 10^{-19}) \approx 1.13 \times 10^{-8}$ cm $\approx 1.23 \times 10^{-7}$ cm (1 cm = 10^7 nm).

Correct answer: G. 1.23×10^{-7} cm

</think>
<answer>
G. 1.23×10^{-7} cm
</answer>